

MOD 301 Course Outline

MOD 301 — Social Accounting Matrix Construction

Course Outline

School	03 Economic Modelling
Level	300 (practitioner / intermediate to advanced)
Hours	35 (indicative band 30–40)
Shape	5 parts · 14 sessions · 7 laboratories · 1 assignment · 1 examination · 1 final project
Model and data	CDCGE — AICTEM Country-Disaggregated CGE (ENVISAGE 10.01 structure) · GTAP 11 (v11b), reference year 2017, values in USD million
Delivery	Instructor-led online, blended, or intensive in-person
Prerequisites	EDIS 101. Intermediate microeconomics and national accounting. Comfort with matrix notation. Julia is used throughout; DEV 801 is recommended but the laboratories are written to be followed without it.
Assessment	Pass 60 %, with 60 % laboratory competency
Version	1.0 · reviewed 2026-08-27 · 12-month cycle

What this course is

A social accounting matrix is the dataset every economy-wide model is calibrated on. Get it wrong and the model is precise about the wrong economy. This course builds one, from the GTAP 11 global database to a balanced, aggregated, model-ready matrix, using the same code that calibrates the AICTEM CDCGE — and spends as much time on the diagnostics as on the construction, because a SAM that balances is not the same thing as a SAM that is right.

Every number in these materials came out of a run. The matrix is Utopia's, built from GTAP 11 for reference year 2017 by the same code that calibrates the AICTEM CDCGE, balanced, aggregated and calibrated until the model reproduced it. Where the build had to assume something the data did not say, the course says which assumption and where: 51 findings are registered against the school, and the appendix carries all of them with the bearing each has on this course.

The rule the course is built on

A SAM that balances is not the same thing as a SAM that is right. Balancing is the easy half. A residual absorbed by RAS is indistinguishable in the finished matrix from a residual that was understood, and the difference between them is the whole of the analyst's contribution.

This course spends five sessions and three laboratories on the part that happens before the balancing routine runs.

Who it is for

Economists, modellers and statisticians who will build or maintain a national SAM, and analysts who must judge whether someone else's SAM can carry the question being asked of it.

Structure

Part	Title	Sessions	Laboratory	Hours
P1	The matrix, its accounts and its naming	01–02	01	4.0
P2	Where the numbers come from	03–06	02	7.5
P3	Diagnosis and balancing	07–10	03–04	10.0
P4	From matrix to model	11–13	05–07	8.8
P5	Documentation and handover	14	—	0.8
	Assignment and assessment time			4.0
	Total	14	7	35.0

The 14 sessions and 7 laboratories, in teaching order, and the subject matter of each:

P1 The matrix, its accounts and its naming — Sessions 01–02 · 4.0 h · Laboratory 01

- **01 What a SAM Is, and What It Is For** — The circular flow as a square matrix; Row = receipt, column = outlay; The six account classes; Why the matrix must be square and balanced; What a SAM is not: not a model, not a forecast; The Utopia benchmark in one table
- **02 Accounts, Sets and Naming** — Activities and commodities: why both; Factors: labour, capital, land, natural resources; Institutions: households, government, capital account; Rest of world; CDCGE's SEC/FAC/HH/GOV/INV/PTAX naming; Reading the account census
- *Lab 01 Read a SAM* — What is in this matrix, and what does each cell mean?

P2 Where the numbers come from — Sessions 03–06 · 7.5 h · Laboratory 02

- **03 Supply, Use and the Make Matrix** — Supply and use tables; The make matrix and secondary production; Commodity-technology and industry-technology assumptions; Why GTAP carries one SEC account per sector; Basic, producer and purchaser prices; Margins and how they move
- **04 Factor Incomes and the Value-Added Block** — Compensation of employees and the skill split; Gross operating surplus and its decomposition; Land and natural-resource rents; Mixed income and the informal sector problem; Factor income to institutions; The Utopia factor block, read line by line
- **05 Taxes, Subsidies and Transfers** — Production, indirect and direct taxes; Import tariffs and export taxes; Factor-use taxes; Subsidies as negative taxes and what that does to RAS; Transfers between institutions; Which instruments the model can actually move
- **06 The Rest of the World** — Exports and imports by commodity; The trade balance as a SAM cell; Foreign saving and what it forces; Bilateral versus aggregated RoW; International transport margins: the hardest account; Why the Utopia imbalance lives in ROW
- *Lab 02 Build a SAM from GTAP* — Can I build a balanced SAM for a country of my choosing, and say what the builder assumed?

P3 Diagnosis and balancing — Sessions 07–10 · 10.0 h · Laboratory 03–04

- **07 Balance Diagnostics** — Absolute and relative balance tests; The imbalance profile by account class; Reading the sign pattern; Residual versus error; The Utopia raw SAM: 14 accounts out of 148; When to fix the source instead of the matrix
- **08 RAS** — The RAS algorithm from first principles; Convergence and the scale-invariant criterion; Choosing the target: row, column or midpoint; What RAS preserves: signs, zeros, proportions; What RAS cannot do: negatives, partial information; The Utopia balance in 3.79 seconds
- **09 GRAS and Negative Entries** — Why a negative cell breaks RAS; The positive-negative split; The quadratic root and the reciprocal scaling; Junius-Oosterhaven and the Lenzen correction; Checking GRAS reduces to RAS on a non-negative matrix
- **10 Cross-Entropy Estimation** — The Kullback-Leibler objective; Row and column totals as constraints; Why the pure-margin case is exactly RAS; Adding information: known aggregates, estimated rates, bounds; Priors, and the honest treatment of a zero; Golan-Judge-Robinson and Robinson-Cattaneo-El-Said
- *Lab 03 Diagnose an Imbalance* — Where is this SAM wrong, and is it an error or a residual?
- *Lab 04 RAS, GRAS and Cross-Entropy* — Do the three balancing methods agree, and where do they part company?

P4 From matrix to model — Sessions 11–13 · 8.8 h · Laboratory 05–07

- **11 Aggregation** — The GTAP 65 commodities; The ENVISAGE coarse-9, medium-30 and fine-63 schemes; What aggregation destroys; Aggregation bias in a CGE; Choosing resolution from the policy question; The beverages-and-tobacco example
- **12 Validation** — GDP three ways from the SAM; Comparison with the national accounts; Comparison with the balance of payments; Ratio and share plausibility screens; Calibration as the final test; The benchmark replication requirement
- **13 Model-Ready Export and Calibration** — What `calibrate_sam` infers and what it assumes; The Calib object, field by field; Square MCP: variables, equations and why they must be equal; Benchmark replication to 1e-10; What a failed replication tells you about the SAM; The zero-iteration solve
- *Lab 05 Aggregate and Prove It Is Safe* — Does my aggregation still support the policy question?
- *Lab 06 Validate Against the Accounts* — Does this SAM agree with what the national accounts say?
- *Lab 07 Calibrate and Replicate* — Does a model calibrated on this SAM reproduce it?

P5 Documentation and handover — Sessions 14 · 0.8 h

- **14 Documentation and Handover** — The metadata a SAM must carry; Vintages, and never mixing them silently; Recording every adjustment; Declaring known defects; Versioning and reproducibility; The SAM technical note template

What a graduate can do

1. Read a social accounting matrix: name every account, say which transaction each cell records, and state the identity each row-column pair enforces.
2. Build a SAM for any of the 47 GTAP regions from the supply-use and trade headers, and say what the builder assumed where the data were silent.
3. Diagnose an unbalanced SAM: locate the imbalance by account class, distinguish a data error from a residual, and choose the fix rather than letting the balancing routine absorb it.
4. Balance a SAM by RAS, GRAS and minimum cross-entropy; state what each method assumes, and show why RAS is the cross-entropy estimator when the margins are the only information.
5. Aggregate a SAM to a working classification and prove that aggregation destroyed nothing that

the intended question needs.

6. Validate a SAM against the national accounts and against the model that will consume it, and report the residuals rather than hiding them.
7. Export a model-ready SAM to CDCGE, calibrate it, and demonstrate that the calibrated model reproduces the benchmark.
8. Write the SAM documentation a second analyst could rebuild the matrix from: sources, vintages, treatments, adjustments and known defects.

The formal learning outcomes, and the map from each to the assessment that tests it, are in 12_LMS/learning_outcomes.md.

Assessment

Component	Weight	What it is
Knowledge quizzes	—	A bank of 112 items, used for formative practice and drawn on by the theory paper, which carries the marks
Laboratory exercises	35 %	7 laboratories, equally weighted. A laboratory is not complete until its validation checks have been read and its named deliverable produced: an irreproducible result is no result
Applied assignments	20 %	SAM technical note
Theory examination	20 %	Closed-book, 90 minutes, drawn from the quiz bank and the review questions
Final project	25 %	A complete SAM build for a country of the learner's choosing: built, diagnosed, balanced, aggregated, validated, calibrated, replicated and documented

Pass 60 % overall and 60 % laboratory competency. Merit 70 %. Distinction 80 %. One resubmission of a single failed component within eight weeks. A laboratory is not complete until every validation check has been read and the provenance of every figure recorded: an irreproducible result is no result.

Data and access

No learner touches restricted microdata. The database is GTAP 11 under the licence held for the Academy, and what ships with the course is the built Utopia matrix and its diagnostics, not the GTAP source. Learners building a second country do so against the same licensed release. Every figure in the materials carries the resolution it was computed at, because a nine-sector and a thirty-sector answer to the same question are different answers.

Materials and set-up

A 200-page Technical Manual, 15 lecture decks of 226 slides, 7 laboratory decks of 77 slides, a Laboratory Manual, Instructor Guide and Learner Workbook, a 112-item quiz bank, 5 assessment files, 11 datasets, 4 code exercises, 3 case studies and the LMS package. The README lists every file, and COURSE_BUILD_REPORT.md records how each was produced.

Before the first laboratory, learners need CDCGE instantiated at D:/EDIP/models/CDCGE and the shared Julia code on the include path. On this host an Application-Control policy blocks native package images, so every Julia command in this course passes `-pkgimages=no`; without it precompilation fails with a message that names neither Julia nor the package. That is school gap M-02, and the full sequence is in the course README and the Laboratory Manual's pre-flight.

The 51 findings registered against this school, and the bearing each has on this course, are carried in the Technical Manual's gap appendix and listed in `COURSE_VERSION.json`.

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